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Abstract

This study sought to investigate post-game hamstring strength recovery of 26 Australian Football League (AFL) players with a previous hamstring strain injury (HSI) across an AFL season. Maximal unilateral isometric knee flexion strength was assessed using an externally fixed dynamometer, and inter-session reliability was measured during the pre-season period. Linear mixed effects models investigated the influence of numerous variables on post-game hamstring strength decrement (relative change between initial weekly test and individual baseline) and individual within-week strength change following gameplay. The test demonstrated good inter-tester reliability (ICC=0.81 - 0.88; CV=6.73 – 7.33), and an acceptable level of error (MAE=5.77 – 7.14%). Player as a random effect strongly influenced post-game strength decrement and within-week strength change (marginal R^2=0.185 – 0.407; conditional R^2=0.455 – 0.654). Within-week hamstring strength change was strongly determined by post-game strength decrement (estimates=0.51 - -0.69; η^2=0.32 – 0.33; both P=<0.001), the interaction between post-game strength decrement and number of days post-game (estimate=0.43, 95% CI=0.20 – 0.66; η^2=0.096; P=<0.001), and by increased relative sprint distance (>75%) (estimate=2.71; 95% CI=0.39 – 5.03; η^2=0.05; P=0.022). This study shows the importance of early individual assessment of post-game hamstring strength in players with prior hamstring injury and could be valuable to inform post-game hamstring recovery in future applications.

Keywords: Australian Football; Hamstring Strength; Isometric Knee Flexion Test

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Introduction

Hamstring injury prevalence in the Australian Football League (AFL) has been a concern for clubs over many years (Freeman et al., 2021). In fact, hamstring strain injuries (HSI) are by far the most prevalent of all injuries in the AFL, with 4.94 new cases reported per club in the 2021 competitive season resulting in players missing 20 matches per club (AFL, 2022). This can prove detrimental to team success as player availability is one of the most important factors in predicting team success (Fahey-Gilmour et al., 2019).

Reducing HSI can be challenging for team medical and conditioning staff due to the possible interaction of many risk factors associated with HSI. Importantly, non-modifiable factors such as older-aged athletes and previous HSI are identified as the strongest drivers of HSI risk (Green et al., 2020). Accordingly, players exhibiting these risk factors should be closely monitored in an attempt to reduce HSI prevalence. Furthermore, modifiable risk factors have also been reported to significantly influence HSI risk, such as increased high-speed running exposure, shorter biceps femoris fascicle length, reduced hamstring strength and unilateral strength asymmetries (Opar et al., 2015; Duhig et al., 2016; Green et al., 2020).

Interestingly, hamstring strength testing has become a popular method of indicating HSI risk in athletes (Green et al., 2020). However, many common hamstring testing protocols place athletes in body positions with limited applicability to sport performance (i.e., seated with 90-degrees hip and knee flexion) (Kollock et al., 2010; Martins et al., 2016). Accordingly, Wollin et al. (2015) developed an isometric knee flexion testing method that positions athletes in a more functional leg position relative to that seen in the terminal swing phase of the running gait cycle (45-degrees hip flexion, 30-degress knee flexion) (Chumanov et al., 2012). This movement range is proposed as the critical point where HSI are likely to occur during high-speed running (Chumanov et al., 2012), a common mechanism of HSI in Australian Football (AF) (Opar et al., 2015). This test uses an externally fixed dynamometer to assess unilateral maximal voluntary isometric knee flexion, and has reported good inter-tester, intra-day (ICC=0.87; 95% CI=0.75 - 0.93) and good inter-week, intra-tester (ICC=0.86; 95% CI=0.74 - 0.93) reliability in elite-youth soccer players (n=16) (Wollin et al., 2015).

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When investigated in elite-youth soccer and sub-elite AF, the isometric knee flexion test developed by Wollin and colleagues (2015) demonstrated a significant strength reduction up to 26h post-match (Wollin et al., 2017; Charlton et al., 2018). However, previous research has identified the importance of weekly strength testing throughout an AFL season as week-to-week variation in game and training loads strongly influences hamstring strength, and therefore, HSI risk (Rogalski et al., 2013). Interestingly, the same testing method used as a secondary injury prevention screening tool reduced HSI incidence by 0.75 cases per 1000h of exposure across a 5-year period in elite-youth soccer (Wollin et al., 2020). While this demonstrates the potential usefulness of this testing method as an on-going hamstring screening tool in soccer, it has not yet been investigated in an elite-AF setting, across an entire AFL season.

Accordingly, our study had two aims: (1) to investigate the inter-session reliability of the isometric hamstring strength test designed by Wollin et al. (2015) in an AFL setting using a high-risk HSI sample; and (2) to determine the impact of elite-AF match play on hamstring strength decrement and within-week change of a high-risk HSI group in the days following a game using the isometric hamstring strength test.

Methods

Experimental Overview

This study was conducted in two parts. Part A occurred in the 2021 pre-season period of an AFL club and investigated the inter-session reliability of an externally fixed dynamometer to assess players' unilateral maximal voluntary isometric knee flexion. Three testing sessions (n=3) were conducted across 15 days. Players included in Part A averaged 4 to 5 familiarisation sessions over 6 weeks prior to the reliability trials. Part B of this study investigated post-game hamstring strength decrement and within-week strength change over the 2021 AFL home and away competitive season. Here, isometric hamstring strength was measured at two time points each week: (1) Initial Test (IT) on the medical screening day, and (2) Re-Test (RT) conducted prior to the weekly main training session. Testing was repeated on a total of 16 weeks throughout the 2021 AFL Premiership Season (rounds 2 - 6, 9 - 15, 18 - 20, and 23). Testing points were limited due to COVID-19 quarantine

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restrictions which prevented access to players during certain weeks. As the number of days separating the game and the two testing points could vary each week, hamstring strength was assessed across multiple time frames (+1 to +5 days post-game). Finally, associations between post-game hamstring strength and specific game performance parameters were explored. Such game metrics were collected from the club's database of match performance from GPS output (Vector S7 [10Hz], Catapult Sport Ltd, Australia). Players consent to data capture as part of the club's daily training environment. Human ethics approval for this study was granted by the UWA Human Research Ethics Office (Reference: 2021/ET000987).

Participants

Part A: 16 AFL players (age=21.19 ± 3.41 years, height=188.94 ± 7.50 cm, weight=85.56 ± 8.11 kg). *Part B*: 26 AFL players (age=22.16 ± 3.59 years, height=189.08 ± 7.19 cm, weight=87.12 ± 9.66 kg). Players included in both parts of this study were determined by the club to form a high-risk HSI group. This included players who had sustained a HSI within the previous two seasons. Club medical staff defined HSI according to the injury resulting in at least one missed game or one full week of training. Additionally, players within their first three years at AFL level were also included in this study. Although older age has been established as a strong determinant of HSI risk (Green et al., 2020), younger players demonstrate higher week-to-week testing compliance compared with senior players, and due to the logistical constraints of testing (i.e., time), the club chose to prioritise the inclusion of younger players with a higher overall injury risk compared to senior players (3+ years at AFL level) (Buttifant et al., 2011; Fortington et al., 2016).

Isometric Hamstring Testing

Each testing occurred at a standardised time of day (morning) using the same equipment. Testing procedures were followed as described by Wollin et al. (2015). Using a medical examination bed, participants' body position was standardised by lying prone on a wedge to ensure 45° of hip flexion and 30° of knee flexion. An ankle strap placed 5 cm proximal to the distal point of the lateral malleolus was attached to the ground via a UU-K200 calibrated load cell (200kgf capacity; 2.0mV/V±1% rated output) (DACELL Co. Ltd., South Korea), which was converted using a DSCUSB strain gauge to USB converter (200

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samples/s at 18 bits noise free) (Mantracourt Electronics Ltd., USA). Participants were instructed to complete two submaximal practice efforts of the test at ~50% and ~75% of perceived maximal effort, respectively, which required them to attempt unilateral knee flexion against the resistance of the chain. Each practice effort was separated by 10sec. A 20sec rest period followed the final practice repetition, before a single maximal voluntary isometric contraction was conducted. All repetitions were 4sec in duration, with instructions of "get ready, pull, pull, pull, and rest" being given. Testing for each athlete took ~2min in total to complete. Raw peak hamstring force (N) was automatically displayed on a live force-time trace on accompanying software (DSCUSB Toolkit, Mantracourt Electronics Ltd., USA), which was not visible to athletes during any trial. Peak hamstring force (N) data was stored in an Athlete Management System (Smartabase, Fusion Sport, Brisbane, Australia).

Data Processing and Statistical Analysis

Part A: Inter-session reliability was determined using the typical error (TE) and coefficient of variation (CV%), which was considered acceptable based on previous research (<10%) (Cormack et al., 2008). The intra-class coefficient (ICC) was calculated and expressed in absolute terms with the following descriptors: >0.9 excellent, 0.8-0.9 good, 0.7-0.8 moderate, <0.7 poor (Meyers & Blesh, 1962). Data analysis was undertaken using a Microsoft Excel spreadsheet (Hopkins, 2017).

Part B: Inclusion in the Part B analysis required that players must have been tested pre-training on both the IT and RT within the week following gameplay. As a result, 21 of the 26 players evaluated in this study were included in the Part B analysis. Three linear mixed effects models were calculated: (1) post-game strength decrement, (2) influence of testing days and strength decrement on within-week strength change, and (3) predicting within-week strength change. Formulas defining both outcome variables can be found below.

$$Strength\ Decrement\ (\%) = \frac{Initial\ Test\ (N)}{Baseline\ (N)\ (Median\ of\ 5\ Highest\ Tests\ Scores\ During\ Pre-Season\ in\ Absence\ of\ Games)}$$

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$$Within - Week\ Strength\ Change\ (\%) = \frac{Initial\ Test\ (N) - Retest(N)}{Initial\ Test\ (N)}$$

The first mixed-effects model attempted to determine the effect of numerous relevant variables on post-game hamstring strength decrement while controlling for player as a random effect. Variables included were the number of days post-game, age, number of previous HSI (within the previous two seasons), HSI in the current season (Yes/No), dominant kicking leg, game players were coming-off (AFL-travel, AFL-local, WAFL [West Australian Football League]), time on ground, absolute and relative GPS workload (distance, sprint distance [>75%] and top-end sprint distance [>85%] of individual maximal speed) and acceleration measures (rapid accelerations >3m/s/h), as well as relevant game statistics (number of contested possessions and tackles per player per game). Relative GPS metrics refer to the difference between a player's GPS game output with their individual season cumulative average for each metric. Total distance and total acceleration load metrics were excluded from the analysis due to large collinearity with time on ground (Pearson's R=0.87 – 0.89; both P=<0.001). Top-end sprint distance (>85%) occurs sporadically in games and due to limited data, the cumulative averages calculated for top-end sprint were unreliable and strongly skewed; therefore, relative top-end sprint was excluded from analysis. The coming-off variable refers to the level of competition of the previous game, and if inter-state travel was involved.

The second mixed effects model assessed the influence that the number of days post-match, the number of days between testing sessions, strength decrement, and the interaction between strength decrement and the number of days post-match, has on within-week strength change, while controlling for player as a random effect. Finally, the third mixed effects model attempted to predict within-week strength change by evaluating all variables included in model 1, with the addition of the number of days between testing points.

Data processing was conducted using R-studio (v.1.4.1717, R Studio Team, 2021; R Core Team, 2021). Specifically, the *tidyverse* suite of R packages was used to wrangle data (Wickham et al., 2019). Numeric variables were standardised using the *scale* function to ensure model fit and convergence. All linear mixed effects models were calculated using

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the *lmerTest* package (Alexandra et al., 2017) and the *lmer* function in the *lme4* package (Bates et al., 2015). Model performance was determined by calculating the mean absolute error (MAE) and root mean squared error (RMSE). Model performance was compared with estimated baseline performance, calculated as the difference in mean strength decrement (model 1) or mean within-week strength change (models 2 and 3) for all players. Model fit was determined by the marginal R^2, conditional R^2 and the Akaike information criterion (AIC). Random effect structures were evaluated by observing the residual variance, random effect variance and intraclass coefficient (ICC). All model fit and random effect structure variables were displayed using the *tab_model* function in the *sjPlot* package (Lüdecke, 2021). Partial eta-squared (η^2) was calculated using the *effectsize* package to determine the effect sizes for variables in each model, and was classified with the following descriptors: ≤0.01 no effect, 0.02-0.05 small effect, 0.06-0.13 moderate effect, ≥0.14 large effect (Ben-Shachar et al., 2020).

Results

Part A

The results of the inter-session reliability assessment are displayed in Table 1. The isometric knee flexion strength test demonstrated a moderate typical error, a good level of absolute test-retest reliability, and an acceptable coefficient of variation (<10%).

Part B

Summary statistics for included variables are displayed in Table 2. The performance of baseline models are displayed in Table 3. The results of the three linear mixed effects models analysed in *Part B* of this study are displayed in Table 4 and Table 5. Effect sizes for variables included in *Part B* can be found in Table 6.

Post-Game Strength Decrement

This model reflects a level of error (MAE=7.14%) significantly smaller than the baseline model (MAE=12.00%). The fixed effects variables demonstrate limited influence on post-game strength decrement (marginal R^2=0.185); however, a substantial amount of variance in decrement is explained by the inclusion of player as a random effect (conditional R^2=0.587). Players accumulating a high number of contested possessions may experience a

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less severe strength decrement (estimate=3.86; 95% CI=0.60 – 7.11; η^2=0.05; P=0.02). Additionally, while not demonstrating significant effects (P>0.05) in the mixed effects model output, age had a moderate effect size (η^2=0.06). As well, small effect sizes were found for total number of HSI and relative sprint distance (η^2=0.02 - 0.03, respectively).

Influence of Testing Days and Strength Decrement on Within-Week Strength Change

The level of error in this model (MAE=6.78%) is slightly lower than baseline model performance (MAE=8.87%). A modest portion of within-week strength change was influenced by the fixed effects alone (marginal R^2=0.371). Again, the inclusion of player as a random effect influenced within-week strength change (conditional R^2=0.455). Post-game strength decrement was found to have a significant influence on within-week strength change (estimate=-0.69, 95% CI=-0.89 - -0.49; η^2=0.331; P=<0.001), as did the interaction between post-game strength decrement and the number of days post-game (estimate=0.43, 95% CI=0.20 – 0.66; η^2=0.096; P=<0.001).

Predicting Within-Week Strength Change

This models level of error (MAE=5.77%) is moderately lower compared to baseline performance (MAE=8.87%). Model fit variables demonstrate a moderate influence of fixed effects (marginal R^2=0.407), and a strong ability to determine within-week strength change when combined with the random effect of player (conditional R^2=0.654). Post-game strength decrement showed a very large impact on within-week strength change (estimate=-0.51, 95% CI=-0.66 - -0.36; η^2=0.32; P=<0.001). Additionally, players coming-off an interstate AFL game (AFL-travel) are likely to record a 7.62% larger within-week strength change compared to playing a local AFL game (estimate=7.62, 95% CI=0.08 – 14.44; η^2=0.07; P=0.029). Finally, higher in-game relative sprint distance is likely to result in a greater within-week strength change (estimate=2.71; 95% CI=0.39 – 5.03; η^2=0.05; P=0.022).

Discussion

Overall, the key outcomes of this study showed the isometric hamstring strength testing method produced a good level of inter-session reliability and an acceptable level of error during an AFL pre-season. This testing method should be considered a reliable option to

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determine hamstring strength in an elite-AF cohort in future practice or research. Secondly, our study demonstrates the complexity of match-related factors on post-game hamstring strength, with large individual differences in response following game exposure, likely due to differing game demands and athlete characteristics. Consequently, we conclude that hamstring strength changes in the days following gameplay are unique to the individual and should be assessed accordingly.

Measuring post-game hamstring strength has become a candidate variable for determining hamstring recovery, as low hamstring strength has previously been identified as a strong risk factor of HSI (Green et al., 2020). Accordingly, the current study found that greater post-game hamstring strength decrement leads to a significant and positive increase in within-week strength change (models 2 and 3). This was expected, since greater strength decrement immediately post-game likely allows for greater relative strength recovery in the days following gameplay. Therefore, post-game hamstring strength decrement could be considered a useful indicator of recovery time. Current practices in elite-AF often consider the number of days since the previous game as a key variable when planning training loads to allow enough time for sufficient musculoskeletal recovery (Freeman et al., 2021). Our data shows that the number of days post-game alone had little influence on within-week strength change. However, when combined with post-game strength decrement, the number of days post-game had a strong interaction effect on within-week strength change. Importantly, this highlights that players with a severe hamstring strength decrement multiple days post-game (≥2) will have a reduced capacity to improve strength later in the week (3-5 days post-game). Therefore, early testing and recognition of hamstring strength decrement could be valuable to inform medical staff of players at elevated risk of HSI, allowing maximum time to implement recovery intervention and training modification.

This study also investigated game-related factors and athlete characteristics that may influence post-game strength decrement. Our findings showed that more contested possessions was related to lower hamstring strength decrement. This outcome, in isolation, was unexpected, however, it is possible that a higher number of contested possessions may be indicative of a more congested game featuring less sprint distance that would tax the post-game function of the hamstring. Interestingly, sprinting load has been acknowledged

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as a key determinant for hamstring fatigue, and therefore, HSI risk (Duhig et al., 2016; Freeman et al., 2021), which means a more congested game might mitigate this risk. However, our strength decrement model could not support this. Our findings showed that higher relative sprint distance displayed a small, non-significant, improvement on post-game strength decrement, as opposed to the expected reduction of post-game strength decrement. Importantly, this variable had a small effect size on post-game hamstring strength decrement and was not statistically significant. Therefore, future research should attempt to clarify the effect of relative sprint distance on post-game hamstring strength and determine if an association exists between contested possessions and sprinting.

Interestingly, age and previous HSI in the past two seasons did not impact post-game hamstring strength decrement. While prior research has evaluated older age and previous HSI as the strongest risk factors for future HSI occurrence (Green et al., 2020; Opar et al., 2021b), within our dataset these variables had only a small influence. These findings likely reflect our selection criteria of players who sustained a HSI in the previous two seasons or were within their first three years at AFL level, with players resembling a younger age demographic (median age=21). As such, it was difficult to determine any influence of previous HSI or age on hamstring recovery. Overall, while previous research suggests that age and prior HSI have a strong influence on post-game hamstring strength recovery, more research across an entire AFL team is needed to more closely investigate this.

Finally, our third model showed that coming-off an interstate AFL game reflects a greater within-week hamstring strength change and a more severe post-game strength decrement compared with AFL home games. This indicates that the hamstrings likely experience delayed recovery following an AFL game with travel, potentially related to player fatigue. However, our results lack confidence, as coming-off an AFL game with travel only displayed a significant relationship to within-week strength change, not strength decrement. Despite this, we expect that this relationship would strengthen with a larger sample in future research. Interestingly, previous literature investigating HSI risk factors rarely include the influence of travel on hamstring strength (Arnason et al., 2004; Green et al., 2020). Player fatigue could likely be a determinant of HSI risk, with lack of sleep

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previously being shown to increase injury risk in athletes (Milewski et al., 2014). However, research is needed to explore such a link with HSI risk.

The current study is not without limitation. Data was only collected on 16 of 23 weeks during the 2021 AFL Premiership season due to COVID-19 interruptions, limiting the number of usable data collection points. Furthermore, due to strict testing criteria and an immovable testing set-up, the three reliability tests were conducted across 15 days in the pre-season, as opposed to the desired 5 or 8 days. Nevertheless, good levels of reliability were still found, and as AFL teams are likely to only test once per week during the pre-season phase, these findings support the tests usefulness in a practical application. Finally, club-determined selection criteria creating a high-risk HSI group limited the application of these findings to compare across an entire AFL team. However, it should be considered that testing all players of an AFL team would be logistically challenging due to time restrictions pre-training. While the high-risk HSI group provided valuable information on the recovery profile of players most susceptible to HSI, it was not reflective of an entire elite AF team's characteristic. As such, it is plausible that hamstring strength may behave differently in a high-risk HSI group, as research has shown that hamstring strength is potentially impaired for up to three years in athletes with a previous HSI (Charlton et al., 2018).

In summary, we confirm that the hamstring strength testing method designed by Wollin et al. (2015) can produce good levels of inter-session reliability in an elite-level AF cohort with an acceptable level of error. Overall, our data reflects the importance of individual-based strength assessment and identifies the potential value in the early assessment of post-game hamstring strength decrement to indicate hamstring recovery needs. Further investigation is warranted to observe post-game hamstring strength in an entire AFL team, across more frequent testing points, to better understand the variables that influence hamstring strength recovery.

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Declaration of Interest Statement

The Authors report no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author, TS, upon request.

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Table 1. Inter-Session Reliability of the Isometric Knee Flexion Strength Test.

		Mean ± SD (N)			TE (95% CI)	ICC (95% CI)	CV%
		Test 1	Test 2	Test 3			
Leg	Left	400.0 ± 62.0	403.9 ± 75.7	408.8 ± 78.0	27.19 (22.18 - 36.35)	0.88 (0.77 - 0.95)	6.73
	Right	395.3 ± 51.3	413.9 ± 74.8	424.9 ± 62.0	30.16 (24.6 - 40.33)	0.81 (0.64 - 0.91)	7.33

Abbreviations, N: newtons of force; SD: standard deviation; TE: typical error; ICC: intraclass coefficient; CI: confidence interval; CV: coefficient of variation.

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Table 2. Summary Statistics of Variables

Variable	Values	N	Minimum	Median	Maximum
IT (N)		139	241	368	541
RT (N)		139	259	380	560
Within-Week Strength Change (%)		139	-16.76	4.49	42.37
Strength Decrement (%)		139	-50.99	-5.35	35.35
Age		139	18	21	33
Number of HSI		139	0	0	4
HSI in 2021?	*No*	111			
	Yes	28			
Dominant Kicking Leg	*No*	68			
	Yes	71			
Coming-off	*AFL (local)*	35			
	AFL (travel)	42			
	WAFL	62			
Time on Ground (min)	*Absolute*	139	10.1	99.1	123
	Relative	139	-8.53	0.02	0.19
Sprint Distance (>75%)	*Absolute*	139	0	145	461
	Relative	137	-5.22	0	0.70
Top End Sprint Distance (>85%)	*Absolute*	139	0	18.1	149.2
Rapid Accelerations (>3m/s/h)	*Absolute*	117	2.8	9.8	24.8
	Relative	117	-1.96	0	0.36
Game Statistics (count)	*Contested Possessions*	139	0	6	21
	Tackles	139	0	2	9

Abbreviations, IT: initial test; RT: retest; HSI: hamstring strain injury; AFL: Australian Football League; WAFL: Western Australian Football League; N: Newtons.

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Table 3. Baseline Model Performance

	Mean Absolute Error (MAE)	**Root Mean Square Error (RMSE)**
Strength Decrement (%)	12.0%	15.4%
Within-Week Strength Change (%)	8.87%	11.4%

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Table 4. Model Fit, Performance, and Random Effect Structure of Mixed Effects Models.

		Post-game Strength Decrement (%)	**Influence of Testing Days and Strength Decrement on Within-Week Strength Change (%)**	**Predicting Within-Week Strength Change (%)**
Model Fit	Observations	115	139	115
	Marginal R^2	0.185	0.371	0.407
	Conditional R^2	0.587	0.455	0.654
	AIC	870.496	1013.371	812.74
Model Performance	MAE	7.14	6.55	5.77
	RMSE	9.19	8.19	6.99
Random Effect Structures	Residual variance ($\sigma^{2)}$	114.09	75.05	67.15
	Random effect variance ($\tau 00$)	111.24_{player}	11.64_{player}	55.46_{player}
	ICC	0.49	0.13	0.36
	Number (RE)	20_{player}	21_{player}	20_{player}

Abbreviations, MAE: mean absolute error; RMSE: root mean squared error; ICC: intraclass coefficient; RE: random effect.

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Table 5. Fixed Effect Variables in Mixed Effect Model.

Model (n)		**Post-game Strength Decrement (%) (n=115)**					**Influence of Testing Days and Strength Decrement on Within-Week Strength Change (%) (n=139)**					**Predicting Within-Week Strength Change (%) (n=115)**				
Variable	*Values*	*Estimates*	*95% Confidence Intervals*			*p-value*	*Estimates*	*95% Confidence Intervals*			*p-value*	*Estimates*	*95% Confidence Intervals*			*p-value*
(Intercept)		-10.72	-19.88	-	-1.55	**0.022**	0.63	-5.43	-	6.69	0.839	-1.69	-12.63	-	9.24	0.762
Days Post-game IT	*1 (reference)*															
	≥ 2	5	-3.41	-	13.41	0.244	1.37	-3.47	-	6.21	0.579	2.95	-3.76	-	9.67	0.389
Days Between IT & RT	*1 (reference)*															
	2						1.81	-2.37	-	5.98	0.396	-1.03	-8.87	-	6.82	0.798
	3						1.30	-3.35	-	5.96	0.583	-1.86	-10.53	-	6.82	0.674
Strength Decrement							-0.69	-0.89	-	-0.49	**<0.001**	-0.51	-0.66	-	-0.36	**<0.001**
Days Post-game IT [≥2] * Decrement							0.43	0.20	-	0.66	**<0.001**					
Age		-3.96	-11.79	-	3.88	0.322						0.42	-5.01	-	5.86	0.879
HSI in 2021?	*No (reference)*															
	Yes	3.47	-5.76	-	12.7	0.461						5.24	-1.7	-	12.18	0.139
Number of HSI		2.26	-0.82	-	5.33	0.15						1.9	-0.45	-	4.24	0.112
Dominant Kicking Leg	*No (reference)*															
	Yes	-0.99	-5.76	-	3.79	0.685						-0.08	-3.71	-	3.56	0.966
Coming-off	*AFL-local (reference)*															
	AFL-travel	-5.75	-14.6	-	3.09	0.203						7.62	0.8	-	14.44	**0.029**
	WAFL	-0.29	-9.43	-	8.85	0.951						4.57	-2.52	-	11.66	0.207

Abbreviations, IT: initial test; RT: retest; HSI: hamstring strain injury; AFL: Australian Football League; WAFL: West Australian Football League. * signifies interaction.

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Table 5. Fixed Effect Variables in Mixed Effect Model (*continued)*

Model (n)		**Post-game Strength Decrement (%) (n=115)**					**Influence of Testing Days and Strength Decrement on Within-Week Strength Change (%) (n=139)**				**Predicting Within-Week Strength Change (%) (n=115)**				
Variable	*Values*	*Estimates*	*95% Confidence Intervals*			*p-value*	*Estimates*	*95% Confidence Intervals*		*p-value*	*Estimates*	*95% Confidence Intervals*			*p-value*
Sprint distance (>75%)	Absolute	0.57	-5.42	-	6.55	0.852					-0.12	-4.58	-	4.34	0.957
	Relative vs norm	2.28	-0.4	-	4.96	0.095					2.71	0.39	-	5.03	**0.022**
Game Minutes	Absolute	-2.24	-7.64	-	3.15	0.415					-0.56	-4.56	-	3.45	0.785
	Relative vs norm	1.89	-2.36	-	6.15	0.383					-0.35	-3.58	-	2.89	0.833
Rapid Accelerations (>3m/s/h)	Absolute	-2.31	-6.97	-	2.35	0.33					-2.69	-6.18	-	0.79	0.129
	Relative vs norm	2.04	-1.76	-	5.83	0.292					-0.42	-3.33	-	2.48	0.776
Top End Sprint (>85%)	Absolute	0.13	-4.3	-	4.57	0.953					-0.15	-3.56	-	3.27	0.933
Game Statistics (count)	Tackles	-0.82	-3.85	-	2.21	0.595					0.16	-2.17	-	2.49	0.894
	Contested Possessions	3.86	0.60	-	7.11	**0.02**					-2.16	-4.73	-	0.41	0.099

Abbreviations, IT: initial test; RT: retest; HSI: hamstring strain injury; AFL: Australian Football League; WAFL: West Australian Football League.

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Table 6. Effect Sizes of Fixed Effect Variables

	Post-game Strength Decrement (%) (n=115)	**Influence of Testing Days and Decrement on Within-Week Strength Change (%) (n=115)**	**Predicting Within-Week Strength Change (%) (n=115)**
	Partial ETA-squared (η^2)	*Partial ETA-squared (η^2)*	*Partial ETA-squared (η^2)*
Days Post-game IT	0.014	0.002	0.008
Days Between IT & RT		0.006	0.002
Strength Decrement		0.331	0.318
Days Post-game IT * Decrement		0.096	
Age	0.058		0.001
HSI in 2021?	0.006		0.023
Number of HSI	0.021		0.026
Dominant Kicking Leg	0.002		0.000
Coming-off	0.017		0.068
Sprint Distance (>75%)	0.000		0.000
Relative Sprint Distance	0.029		0.053
Game Minutes	0.001		0.001
Relative Game Minutes	0.001		0.001
Rapid Accelerations (>3m/s/h)	0.013		0.037
Relative Rapid Accelerations	0.012		0.001
Top End Sprint (>85%)	0.000		0.000
Tackles	0.003		0.000
Contested Possessions	0.052		0.028

Abbreviations, IT: initial test; RT: retest; HSI: hamstring strain injury; AFL: Australian Football League; WAFL: West Australian Football League. * signifies interaction.

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